

Scaling from Metawebs to Realised Webs: A Hierarchical Approach to Network Ecology

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Abstract: Ecological networks provide a critical framework for understanding the architecture of biodiversity and predicting ecosystem responses to environmental change. However, the application of network ecology is often hindered by a lack of clarity regarding the assumptions inherent in different network representations. While recent literature predominantly focuses on data-centric classifications, predictive missing-link inference, or the topological specification of nodes and links, we focus instead on the theoretical gradients and processes that govern network construction. Here, we present a hierarchical framework that distinguishes between ‘metawebs’ (representing the fundamental feasibility of interactions) and ‘realised webs’ (representing interactions expressed in specific spatiotemporal contexts). We identify five core processes that drive the transition from potential to realised interactions: evolutionary compatibility and co-occurrence, which define the feasibility of links; and abundance, diet choice, and non-trophic interactions, which determine their realisation. Furthermore, we map these processes onto a methodological spectrum of network construction, distinguishing between inductive approaches that infer structure from observation, and deductive approaches that generate structure from mechanistic first principles. By making explicit the underlying biological assumptions and scale-dependent processes, this framework moves the field past descriptive topology, clarifying the scope of inference possible with each approach to facilitate more robust predictions of biodiversity dynamics in the anthropocene.

Keywords: food web, network construction, biodiversity, scale and process, interaction modelling, trophic network

1 Introduction

At the heart of modern biodiversity science are a set of concepts and theories about species richness, stability, and function, which have been discussed since the foundational work of [1], [2], and [3] and more recently [4]. These relate to the abundance, distribution, functions, and services that biodiversity provides. Network representations of interactions among organisms are increasingly argued to be an asset to understanding and predicting the impacts of multiple, simultaneous stress on biodiversity (*e.g.*, foundational studies on food web structure and robustness: [5–8]). Documenting interactions is thus one of the fundamental building blocks of community ecology and provides a powerful abstraction for mathematical and statistical modelling of biodiversity to make predictions, and to mitigate and manage threats [9].

However, there is a growing discourse around limitations to the interpretation and applied use of networks, which have been recognised since early discussions of sampling effort and aggregation in food webs ([5,10,11]; recent discussions: [12,13]). Against this, it is important to evaluate the value and the limitations of the various network conceptualisations and how these relate to biodiversity concepts, such as community structure or ecosystem function [14]. In this perspective we aim to provide an overview of different **food web** representations, particularly how each representation embeds assumptions about the processes that determine interactions (Section 3) about the levels of organisation at which this occurs (*i.e.* the biological, ecological, spatial/temporal scales) and the way in which we construct the resulting networks (Section 4).

Network construction reflects both the data used and the theories governing species interactions. We still lack a clear explanation of the different assumptions and scale dependent processes that underpin network construction alongside extensive discussions about the challenges relating to data collection and observation [6,10,14–20]. Such an understanding should deliver an acceleration in capacity to more effectively predict the impact of multiple stressors on biodiverse communities.

In their recent work, [21] synthesised ecological networks based on node resolution and link type, with a focus on the data and methodologies underlying each representation. Here, we take a complementary theory-driven perspective by framing network construction as a hierarchical transition from feasibility to realisation of interactions. In the following sections, we review how nodes, edges, and interaction processes shape different network representations, highlighting how changes in assumptions alter ecological scale, node and

link resolution, modelling approaches, and the relative importance of evolutionary versus ecological processes. We conclude by aligning these representations with key questions in biodiversity science in the Anthropocene.

2 Setting the Scene: The Not So Basics of Nodes and Edges

Ecological networks serve multiple uses, representing an ‘object’ from which inferences can be made. While many aspects of community structure can be analysed without networks (*e.g.*, trait distributions or abundance patterns) networks provide a formal framework for capturing the organisation among species. The study of network structure and topology has a long history in ecology, rooted in early theory on energy flow [2,22], and later extended to questions of robustness, stability, and complexity (*e.g.*, [7,20,23–25]). More recent work has built on this foundation to link network structure to ecosystem functioning, persistence, and dynamical behaviour (*e.g.*, [26–28]). Networks are therefore commonly treated as response variables in tests of ecological theory and statistical models of the generative processes that give rise to observed structure and are widely used to compare communities across environmental gradients or through time [*e.g.*, 29,30]. They also provide a platform for evaluating downstream responses to perturbations, including secondary extinctions and robustness to species loss (*e.g.*, [7,31–33]), as well as inference about stability, ecosystem function, invasions, climate change, contaminants, and extinction cascades (*e.g.*, [34–36]). Against this backdrop of multiple research agendas, the definition of ‘edges’ and ‘nodes’, and the levels of organisation at which they are defined, take many forms [17,37], each of which encode a series of assumptions within a network. Here we introduce a perspective on these baseline assumptions.

2.1 How do we define a node?

While nodes are conventionally described as representing species, in practice they may correspond to a range of taxonomic and non-taxonomic units, including sub-species, genera, or families, as well as trophic species (*e.g.*, [38,39]), feeding guilds (*e.g.*, [40]), or life-stage-specific subsets of species (*e.g.*, [41]). These choices reflect differences in the level, type, and consistency of resolution at which networks are constructed, rather than aggregation per se. Representing nodes at coarser or mixed resolutions can limit taxon-specific inference (*e.g.*, whether species *a* consumes species *b*), bias estimates of degree distributions (particularly

generality and vulnerability) and complicate downstream analyses such as extinction or invasion dynamics, where species identity and the consequences of loss may be obscured [41,42]. At the same time, there are clear justifications for using aggregated representations when the distribution of interactions among functional or trophic units is more informative than species-level detail, for example when analysing extinction patterns across feeding guilds [43]. More broadly, issues of resolution, scale, and sampling have long been recognised as central to the construction and interpretation of food webs (*e.g.*, [6,44]).

2.2 What is captured by an edge?

Understanding edges requires distinguishing between *potential* and *realised* links - potential links reflect feasibility, whereas realised links reflect fluxes such as energy transfer. Links within food webs are thus a representation of either potential links between species [45] or fluxes within a system *e.g.*, energy transfer or material flow as the result of the feeding links between species [2,46]. Edges can thus correspond to different ‘currencies’ [21]. There is also a myriad of ways in which the links themselves can be specified. Links between species can be treated as present or absent (*i.e.*, binary), may be defined as probabilities [47,48] or by continuous functions which further quantify the strength of an interaction [49]. Link definition depends on both the ecological currency and how interactions are represented. For example, feasibility is unlikely to accommodate flux, but does align with binary or probability representations. Taking a food web that consists of links representing feasible interactions among a collection of species will be meaningless if one is interested in understanding the flow of energy through the network as the links are not environmentally/energetically constrained.

2.3 Network representations

Given these definitions of nodes and edges, ecological networks can be divided into two broad types: **metawebs**, which represent all *potential* interactions within a species pool [44], and **realised networks**, which represent the subset of interactions expressed within a particular community at a given place and time. These representations differ in both the scales at which they are constructed and the processes assumed to generate their structure.

A metaweb is fundamentally a list of *feasible* interactions between species. Feasibility is determined by trait complementarity, typically related to feeding, and can be further

96 constrained by species co-occurrence, producing a transition from *global* to *regional*
97 metawebs. Metawebs therefore identify evolutionarily and regionally plausible interactions,
98 ecologically impossible (*i.e.*, forbidden) links [50], and the potential diet breadth of species
99 [51].

100 In contrast, realised networks are localised in space and time, with links shaped by co-
101 occurrence, environmental conditions, and diet choice. Even when represented as binary
102 matrices, their links implicitly reflect interaction strength and energy transfer. Realised
103 networks are therefore not simple spatial or temporal subsets of metawebs; they emerge
104 from the processes governing whether feasible interactions are expressed. Consequently,
105 a metaweb and realised network containing the same species may differ substantially in
106 structure because link presence is governed by different constraints. Links absent from a
107 metaweb represent infeasible interactions, whereas links absent from a realised network
108 reflect context-dependent ecological constraints.

109 **3 From Nodes and Edges to Process and Constraints**

110 In the previous section we discussed how the definition of nodes and edges, representing
111 different scales and processes, lead to the concept of a metaweb and a realised web. The
112 fundamental take-homes are that nodes vary in their resolution, edges vary in what kind
113 of process they represent and the intersection of these, defined by meta- vs. realised webs,
114 underpins distinct lines of inquiry and constraints on the type of inference we can make
115 with networks. Here we reveal five core constraints across evolutionary and ecological
116 scales that further delineate the transition from meta- to realised webs, exposing processes
117 that determine the nature of links among nodes: evolutionary compatibility, co-occurrence,
118 abundance, diet choice, and non-trophic interactions Figure 1.

119 [Figure 1 about here.]

120 **3.1 Processes that determine the feasibility of an interaction**

121 Evolutionary compatibility and co-occurrence are the two principal processes that defines
122 the feasibility of an interaction between two species. The scale of inference and set
123 of processes embodied in these two constraints typically combine to define a ‘list’ of
124 interactions that are viable/feasible and defined strictly as present/absent. Reflecting on
125 the previous section, nodes are typically species and rules defining edges are defined by
126 trait complementarity (phylogenetic) and/or co-occurrence. Here we provide more insight

127 into each process.

128 **Evolutionary compatibility**

129 This constraint is defined by shared (co)evolutionary history between consumers and
130 resources [52–55] which is manifested as ‘trait complementarity’ between two species [56].
131 In this body of theory, the consumer has the ‘correct’ set of traits that allow it to acquire
132 and consume the resource. Interactions that are not compatible are defined as forbidden
133 links [50]; *i.e.*, they are not physically possible and will *always* be absent within a network.
134 Networks do not properly arise from models based on this constraint. Instead, interacting
135 species pairs are defined and these are represented as binary (possible vs forbidden) or
136 probabilistic [47]. For example, in the metaweb constructed by [57] probabilities are
137 quantified as the confidence of a specific interaction being *possible* between two species. A
138 network constructed based on evolutionary compatibility is conceptually aligned with a
139 ‘global metaweb’, and gives us information as to the global feasibility of links between
140 species pairs despite the fact that they do not co-occur (see Figure 1).

141 **(Co)occurrence**

142 The co-occurrence of species in both time and space is a fundamental requirement for an
143 interaction between two species to occur (at least in terms of feeding links). Although
144 co-occurrence data alone is insufficient for building an accurate and ecologically meaningful
145 representation of *feeding links* [58], it is still a critical process that determines the possible
146 realisation of a feeding. Knowledge on the co-occurrence of species allows us to spatially
147 constrain a global metaweb to reflect regional metawebs [59]. In the context of Figure 1
148 this would be the metawebs for regions one and two.

149 We reinforce that these two constraints don’t deliver a network *per se*, but a list of feasible
150 species pairs. Although it is possible to build a network from the list of interactions
151 generated by these constraints, it is important to be aware that the structure of this
152 network is not constrained by any community context - just because species are able to
153 interact does not mean that they will [60,61].

154 **3.2 Processes that realise networks**

155 In contrast to the above, here we highlight three processes that influence the *realisation* of
156 an interaction between species and thus form the conceptual basis for realised networks.
157 As we show in Figure 1, a ‘truly realised’ network is the product of properties of the

community (**abundance** and **non-trophic interactions**) and the individual (**diet choice**). This represents a conceptual shift from considering the feasibility for species pairwise interactions to considering the edge as a representation of energy flow. Such a transition requires information about how the community, the environment and the individual *constrains* network topology as defined by consumer choice ([62], Section 2.3)

Abundance

Abundance as a realising process emerges from a null model for energy acquisition: organisms feeding randomly will consume resources in proportion to their abundance [63]. Here, abundance of different prey species influences the distribution of links in a network [64] by defining a preference linked to individuals among species meeting [47,60]. Abundance data (linked to a derived metaweb) delivers a foundation ruleset that can define the distribution and strength of links. Of note, however, is that such abundance constrained interactions are not necessarily contingent on there being any compatibility between species [65–67].

Diet choice

It is well established that consumers make more active decisions than eating items in proportion to their abundance [63]. Ultimately, consumer choice is underpinned by an energetic cost-benefit framework centred around profitability and defined by traits associated with acquisition and consumption of a resource [68,69]. Energetic constraints are invoked to construct networks in a myriad of ways [*e.g.*, 42,70–72].

Unlike metaweb approaches, these models generate realised webs as emergent outcomes of consumer behaviour. We also here make a distinction, developed below, with models like the Niche Model [39], where diet choice is implicit in its probabilistic network generating function, but it is working to replicate the *expected* structure of the network, and this structure does not emerge from node-based rules. Note that we select diet choice as a term to capture rules linked to optimal foraging [73] and metabolic theory [74] for capturing the energetic constraints on the distribution and strength of interactions.

Non-trophic interactions

We include non-trophic interactions [see 75] here not as a determinant of links, but a modifier of them - they are the community context above and beyond co-occurrence and abundance. Non-trophic interactions include competition for space, predator interference, refuge provisioning, recruitment facilitation as well as non-trophic effects that increase or

190 decrease mortality. These interactions specifically modify either the realisation or strength
191 of trophic interactions [28,31,76–78] and represent direct (*e.g.*, predator *a* outcompetes
192 predator *b*) and indirect (*e.g.*, mutualistic/facilitative interactions) mechanisms.

193 Some interactions, such as pollination, occupy an intermediate position in this framework,
194 as they combine trophic components (*e.g.*, resource consumption) with non-trophic effects
195 that influence reproduction, recruitment, and population persistence [79–81]. They operate
196 on the realisation of a network by altering the fine-scale distribution and abundance of
197 species and relative contributions of direct and indirect effects to biomass, persistence,
198 stability, and the functioning of the communities [75,82–84].

199 4 Network construction

200 The above five processes are central to understanding the assumptions inherent in building
201 different types of networks. Each of the processes, or combinations thereof, deliver a
202 unique set of boundary conditions on what a network represents and can be used for. Here
203 we build on the introduction of these five processes to further categorise the approaches
204 to constructing networks.

205 4.1 Why construct networks?

206 Networks are a representation of biodiversity. In a perfect world, we might know about
207 all interactions. However, the empirical collection of interaction data is both costly and
208 challenging to execute [50,85,86]. In the absence of robust empirical data, we use ‘models’
209 that facilitate interpolation and gap-filling of existing empirical datasets [*e.g.*, 87,88–90],
210 predict the feasibility of interaction among pairs of species, or directly predict network
211 structure [see 91 for a broader discussion].

212 Networks are unique in delivering more than just estimates of species richness. As noted
213 in the introduction, a network embodies the organising structure of biodiversity and allows
214 numerous opportunities for ‘downstream’ analysis, including the comparison of structures,
215 estimation of energy flux or extinction dynamics and ultimately form the structural
216 inputs to dynamical systems models that facilitate ecological and conservation relevant
217 inference about productivity-diversity-stability-function relationships [27] in space and
218 time. But making such inferences requires careful attention to one or more of the processes
219 discussed in Section 3. While these network representations may be simplifications of
220 the truth [92], they remain critically useful tools as they allow us to move beyond simple

descriptions of species richness to test hypotheses about community architecture and ecosystem functioning that would be otherwise impossible to assess.

4.2 Construction through induction

Constructing feasible or realised networks can be framed as an ‘inductive reasoning’ process where insight and generalisation arise from a set of observations and relationships around feeding. Inductive reasoning as a foundation for network construction is implemented at node and network levels.

4.2.1 Species specific networks: construction through node level induction

When applied at the node level, species-specific networks are created based on expected feeding interactions. When applied at the network level, networks are judged by their structural properties. All methods in this inference space rest on a set of three assumptions: there are a set of ‘feeding rules’ that underpin interaction feasibility [93]; these rules are phylogenetically conserved [54,94]; and they can be specified by matching the traits between consumer and resource.

Evolutionary compatibility and co-occurrence constraints have been critical to the construction of ‘first draft’ networks for communities for which we have no interaction data [57]. They are also central to interpolation in data poor regions and predicting interactions for ‘unobservable’ communities *e.g.*, prehistoric networks [43,95,96] or future, novel community assemblages [97]. Furthermore, they have the capacity to evaluate a role of interactions among species relative to their distribution by accounting for the role of the environment and the role of species interactions [98–101].

Feeding rules can be specified in several ways. Expert-based approaches define feasible interactions using trait matching [102,103], while mechanistic approaches often rely on trait-based relationships between consumers and resources [104,105]. Alternatively, statistical and machine-learning methods infer interaction probabilities from observed networks using ecological predictors such as traits and phylogeny [106].

Rules are also defined by correlating real world interaction data with suitable ecological proxies for which data is more widely available (*e.g.*, traits) using some sort of binary classifier (see [106] for an overview). These include generalised linear models [*e.g.*, 107], random forest [*e.g.*, 108], trait-based k-NN [*e.g.*, 109], and Bayesian models [*e.g.*, 110,111]. Finally, graph embedding uses the structural features of a known network to infer the

position of species in an unknown network through the decomposition of the interaction onto the embedding space (see [51]). This decomposition relies on a combination of ecological proxies (*e.g.*, phylogenetic relatedness [57]) in conjunction with known interactions to infer the latent values of species, which can then be mapped onto decomposition of a known network.

4.2.2 Species agnostic networks: construction through structure induction

These models generate ecologically realistic network structures using simple probabilistic rules. They are commonly used as null expectations, for comparative analyses, and as inputs to dynamical models. The determination of links between species is not directly linked to properties of the nodes. This means these networks are usually not species specific. Although they require little empirical information, they encode explicit assumptions about expected network structure.

Stochastic network models use a probabilistic ruleset about diet choice and niche breadth to reflect fundamental ideas of foraging biology. These models that are based on the compartmentalisation and acquisition of energy for species at different trophic levels [112,113] and that network structure can be determined by distributing interactions along single dimension (the ‘niche axis’; [114]). Typically, these models parametrise some aspect of the network structure (although see [112] for a parameter-free model). These models include the most used network generator, the Niche model [39], as well as the original Cascade model [5] and the derived Nested hierarchy model [115]. Even though these networks are derived without any real-world data they are still able to recover the structure of empirical networks [116]. These models often form the basis for dynamic models *e.g.*, the allometric trophic network [24,26] and bioenergetic food web models [34].

4.3 Construction through deduction

In contrast to the above approaches centred on feasibility, the construction of realised networks relies on deductive reasoning and have a unique agenda to those above. In contrast to the inductive methods, inference about a realised network follows from a set of premises defining generative processes, often referred to as mechanisms, allowing inference about the distribution and strength of interactions. These models are widely used to study energy flux, extinction dynamics, and ecosystem functioning. They also provide the structural backbone for dynamical systems modelling to address questions about stability-structure-productivity-function relationships, secondary extinction dynamics,

species invasion and climate change. There are two broad groups of models in this deductive category.

4.3.1 Species-specific networks

These models capture the behaviour of the nodes by explicitly considering the properties of the different species in the community. Which means that there is a degree of variance in which links are predicted between species unlike the more ‘static’ predictions made by inductive models. However, these networks are costly to construct in real world settings (requiring data about the entire community, as it is the behaviour of the system that determines the behaviour of the part) and also lack the larger diet niche context afforded by metawebs.

Neutral networks are built on the assumption that foraging decisions are tied *only* to the abundance of species within the community [117,118]. Here links are solely determined by the relative abundance of the different species in the community. Although unrealistic as a complete explanation, neutral models can be combined with inductive approaches to generate more localised predictions [67].

There is a broader group of models that focus on determining interactions in terms of energetic constraints on diet breadth, often using the ratio of consumer-resource body size as a proxy for capturing the energetic constraints of feeding. Models such as those developed by [71] and [68] are similar to the mechanistic approaches discussed in Section 4.2, however instead of determining interactions based on mechanistic feasibility it is rather constrained by the energetic cost of predation. Note that although these models do not place any explicit constraints on the expected structure of the network, the links should still be considered as ‘realised’ owing to the energetic constraint placed on links. A different subset of diet models [*e.g.*, 42,119] use a diet choice approach, however like the stochastic network models they also embed assumptions on network structure. Thus, these models predict both interactions and network structure simultaneously, although they would benefit in being refined by more explicitly accounting for trait-based (*i.e.*, feasibility) parameterisation [35].

5 Making Progress with Networks

The motivation to leverage network ecology in conservation ecology, environmental risk assessment and natural resource management stems from a shift away from species/popu-

315 lation specific measures of the effects of stress and disturbance to community level metrics
316 of these impacts. These metrics, such as resilience and more generally stability, ecosystem
317 function and biodiversity, are natural properties of networks. This suggests that modern
318 conservation, risk assessment and resource management requires robust network tools to
319 support decision making.

320 This is also true in the disciplines of ecology and environmental science and their focus
321 on abundance, distribution, functions and services that biodiversity provides [4]. Major
322 questions remain, for example, about stability-diversity-productivity relationships, the
323 impacts of extinctions and invasions, and the impacts of multiple stressors operating at
324 multiple ecological scales. A network approach to answering these types of questions
325 specifically allows us to evaluate how environmental gradients and anthropogenic stress
326 map through direct and indirect effects among species in a complex community and reveal
327 fundamental patterns and understanding of processes in the natural world.

328 To effectively use networks to aid us in answering questions about conservation/risk
329 assessment/management and core ecological theory, we need to be mindful that we are
330 mapping the *correct* network representation to the question of interest [21]. Notably, there
331 are certain questions that cannot be answered using specific network representations as
332 the scale of the question of interest is fundamentally misaligned with the process captured
333 by a specific network representation (Section 3.1), the underlying data that is used to
334 construct it (Section 4), or both.

335 Here we discuss and map the different network representations shown in Figure 1 to
336 ‘appropriate’ research questions and agendas (see also Table 1). We also highlight some of
337 the key methodological challenges that currently limit our conceptualisation of a ‘network’
338 and thus impact their effective practical application in real world settings.

Table 1: Showcasing some of the broader avenues of inquiry, specifically how they map to the different network representations. Additionally, we highlight some studies that address or present opening discussions around each research question. Superscripts at the research question indicate the strength of the current literature in addressing that research question: $\checkmark$ indicates area with strong foundational research, Δ partial/emerging areas of research, and $\times$ areas where research is weak/largely absent

Network		
Representation	Example Research Question	Representative Studies
Global Metaweb	How will novel communities respond to <i>e.g.</i> , extinction, turnover, invasion and rewilding $\checkmark$	[120]; [7]
	Diet-based conservation focusing not only on the target species but the species it might depend on for food resources Δ	[121]; [122]; [123]
	Rewiring capacity/potential of species by looking at the <i>entire</i> diets of species Δ	[124]; [31]; [125]; [126]
	Eco-Evolutionary dynamics and how they relate to the conservation and origination of feeding strategies $\times$	[60]; [127]
Regional Metawebs	Applied use potential of questions highlighted for global metawebs at the management scale <i>e.g.</i> , a protected area Δ	[101]; [128]; [129]
	Refinement/extension of species distribution models by incorporating co-occurrence and species associations <i>e.g.</i> , predator and prey $\checkmark$	[130]; [131]

Network		
Representation	Example Research Question	Representative Studies
Realised Webs	The allocation of multiple stressors across networks Δ	[132]; [133]
	Temperature threshold to community collapse Δ	[134]; [135]
	Extinction and persistence after harvesting/invasion/extinction $\checkmark$	[136]; [137]
	Stability-diversity-productivity-function $\checkmark$	[138]; [121]
	Explicitly tying ecosystem level processes and nutrient flows to networks $\times$	[139]
	Meta communities and the idea of meta-network-communities Δ	[140]; [141]

5.1 Key Eco-Evo-Conservation Questions

5.1.1 Global Metawebs

Global metawebs are most appropriate for questions centred on interaction feasibility and potential diet breadth. They provide a platform for exploring hypothetical or novel communities under climate change [142], species invasions, reintroductions, and rewilding, and the potential rewiring capacity [126]. Because they focus on feasible rather than realised interactions, global metawebs are also well suited to studying eco-evolutionary dynamics and how evolutionary history, natural selection, and phenotypic plasticity shape interaction niches.

5.1.2 Regional Metawebs

Regional metawebs extend these questions by explicitly incorporating species co-occurrence and therefore provide a more management-relevant perspective. They can be used to refine species distribution models and projections of future community composition [29,40]. However, caution is required when comparing regional metaweb structure across space

or environmental gradients, as observed differences may reflect species turnover (*e.g.*, β -diversity) rather than changes in interaction processes.

5.1.3 Realised networks

Realised networks are best suited to questions concerning how community and environmental context shape interaction structure and ecosystem functioning [143]. They provide the appropriate framework for studying stability, resilience, biodiversity dynamics, ecosystem functioning, extinction cascades, invasions, climate change impacts, and network rewiring through time. By capturing interactions that are actually occurring, realised networks allow investigation of how perturbations propagate through communities and influence persistence.

The increasing availability of long-term interaction datasets is expanding opportunities to address these questions [27,144]. However, empirical datasets often accumulate interactions across extended periods, potentially obscuring temporal variation in realised interactions [10]. Developing approaches that better reconcile empirical networks with realised community dynamics therefore remains an important challenge.

5.2 Key methodological challenges

As noted above, the three network types highlight longstanding methodological challenges that limit both the precision and accuracy of ecological inference. Here we briefly review these challenges and emerging opportunities to address them.

Understanding what empirical data represents: Robust inference requires understanding what constitutes an observed interaction, whether recorded directly (predation events) or indirectly (*e.g.*, gut contents or stable isotopes). Because empirical networks often accumulate observations across space and time, they may be conceptually closer to metawebs than realised networks.

The validation of network structure: Considerable progress has been made in assessing how well models recover pairwise interactions [91,145], yet there remains no clear framework for evaluating their ability to recover network structure [114,146]. This raises two related questions: what constitutes an appropriate benchmark, and which aspects of network recovery matter most? For metawebs, accurately identifying both present and forbidden links may be essential, whereas for realised webs it remains unclear whether recovering pairwise interactions, aggregate properties (*e.g.*, connectance), or both should

384 be the primary objective.

385 **Transitioning between metawebs and realised webs:** Most approaches for modelling
386 realised networks do not explicitly incorporate evolutionary constraints (although see
387 [147]; [68]). Progress will likely require either ensemble approaches [148,149] or methods
388 for downscaling metawebs into realised networks [e.g., 150]. However, any such transition
389 must retain clarity about the meaning of links - structurally realistic networks do not
390 necessarily represent realised prey choice or energy flow.

391 Developing frameworks that allow transitioning between metawebs and realised rep-
392 resentations will also facilitate integration with metacommunity, metaecosystem, and
393 ecosystem-level theory [151,152]. Doing so requires clear definitions of what constitutes a
394 network and where its spatial and temporal boundaries lie [153].

395 **Making networks more tractable in applied spaces:** The application of networks
396 to conservation and management remains limited by difficulties in defining network
397 boundaries, aligning them with management units, and interpreting network metrics in
398 policy-relevant ways [154]. Addressing these challenges will require stronger links between
399 network structure and ecosystem function, alongside careful matching of analytical tools
400 and network representations to management objectives [128,155].

401 Taken together, these challenges highlight three overarching messages. (i) Network
402 representations are inseparable from the data and assumptions used to construct them;
403 (ii) validation and benchmarking must be aligned with the intended network type and
404 research question; and (iii) greater conceptual clarity is needed when transitioning between
405 metawebs and realised networks, particularly in applied contexts. Explicitly articulating
406 these distinctions is essential if networks are to be used rigorously and transparently across
407 scales.

408 6 Concluding Remarks

409 1. **Network representations encode different processes:** Different network repre-
410 sentations capture distinct assumptions, data structures, and ecological processes;
411 understanding the interplay between a network's structure and the processes it
412 encodes is critical for determining which questions it can address.

413 2. **Network representations are not interchangeable:** There is no universally
414 optimal network representation. The utility of a network depends entirely on how

well its underlying assumptions and data align with the intended scale and process of analysis.

3. Explicit assumptions are essential for evaluation: Explicitly articulating the assumptions behind network assembly facilitates a more critical evaluation of network suitability and inferential power. It is important that we move beyond the uncritical and default reliance on conventional network representations in favour of understanding the consequences of selecting a specific representation or reconstruction/modelling approach.

4. A unified framework improves application and progress: The framework presented here provides a structured basis for comparing network representations and evaluating their suitability across conceptual, methodological, and applied contexts. Establishing this standard is essential for preventing the misinterpretation of network data and for addressing foundational questions in biodiversity science.

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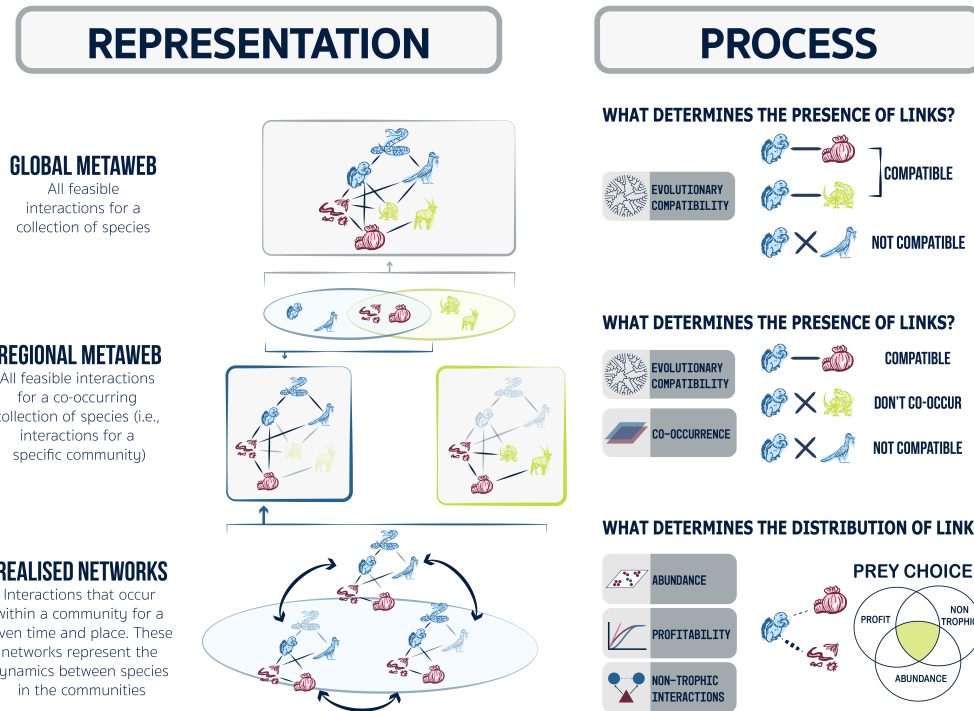


Figure 1: Aligning the processes that determine interactions (right) with different network representations (left). A **global metaweb** captures all feasible interactions among a species pool. Restricting this network by species co-occurrence yields **regional metawebs**, which differ in species composition. Species occurring in both regions are shown in red, whereas region-specific species are shown in blue and yellow. **Realised networks** represent subsets of regional metawebs expressed in specific spatial and temporal contexts. Their structure is shaped not only by co-occurrence, but also by community-level processes such as abundance, diet choice, and non-trophic interactions.